



Smart Helmet for Accident Detection & Rider Safety

Team Members:

- **Zohaib Shah** (22F-BSCS-87)
- **M. Rayyan Naveed** (22F-BSCS-66)
- **Sajid Khalid** (22F-BSCS-76)
- **M. Saim** (22F-BSCS-80)
- **Shayan** (22F-BSCS-93)

Supervisor:

- **Dr Zulfiqar Hussain**

Department & University:

- Final Year Project – BSCS
- **Dawood University of Engineering and Technology (DUET), Karachi**



Smart Helmet Project – Enhancing Road Safety in Pakistan

Our Smart Helmet is designed to improve motorbike rider safety, especially in accident-prone regions like Pakistan. It combines preventive safety measures with emergency response features, making it a practical and life-saving innovation for everyday riders.

Problem Statement

Motorbike riders face high accident risks due to poor safety practices and lack of smart helmets. Traditional helmets cannot detect accidents or send alerts, increasing the chances of severe injuries and fatalities.

Introduction: A Life-Saving Innovation for Road Safety

The **Smart Helmet** is an innovative safety device designed to enhance the protection, awareness, and overall safety of motorcycle riders by integrating modern technologies such as Internet of Things (IoT), Artificial Intelligence (AI), Computer Vision, and Data Analytics. It aims to reduce the rising number of road accidents by ensuring that riders always wear their helmets and by continuously monitoring critical data such as head movement, speed, impact force, and surrounding conditions. The system utilizes sensors like the accelerometer, gyroscope, and GPS to detect accidents, while an AI-driven model helps analyze data in real-time to identify risky behaviors or hazardous situations. In the event of an accident, the helmet automatically sends an alert along with the rider's GPS location to emergency contacts through cloud connectivity. Additionally, the inclusion of computer vision enables features such as obstacle detection, drowsiness detection, and traffic sign recognition. Through data analytics, insights are generated about rider habits, environmental risks, and system performance, making the Smart Helmet not just a safety device, but a complete intelligent monitoring system aimed at saving lives and improving road safety.



Literature Review: Identifying the Gaps in Current Smart Helmet Solutions

Previous research in this area has focused mainly on individual aspects of smart helmets. Some projects used RFID technology to check if the helmet is worn, while others developed systems for accident detection using accelerometers and GSM modules. There have also been attempts to integrate alcohol detection into helmets, but such features were not always practical in every cultural or regional context.

Helmet Safety Systems (2016–2020):

- Many IoT-based helmets detect *helmet wearing* and send alerts via *RFID* or *Bluetooth*.
- Limitation: No real-time *accident detection* or *cloud integration*.

Accident Detection Models:

- Projects used sensors like *MPU6050* or *ADXL345* to detect impacts.
- Limitation: High false positives — sudden braking also triggered alerts.

GSM/GPS-Based Helmets:

- Systems like GSM+GPS Smart Helmets could send location via SMS after impact.
- Limitation: No AI or *machine learning-based accuracy improvement*.

Computer Vision Research:

- Studies using *OpenCV* and *YOLO* for helmet detection were effective in lab setups.
- Limitation: Poor performance under *real-time lighting* or *outdoor conditions*.

This analysis underscores a critical need for a holistic, cost-effective, and highly reliable Smart Helmet system – a need our project is specifically designed to address by integrating diverse functionalities into a cohesive and practical solution.

Research Objectives: Building a Comprehensive Safety System

Our project aims to overcome the limitations of existing smart helmet technologies by delivering a robust, and highly effective safety solution using IOT, AI,CLOUD ,Computer Vision,Data Analytics for motorcycle riders.

01

Objectives:

1. **Helmet Verification System:**

- Detect if the rider is wearing the helmet before allowing bike ignition.

2. **Accident Detection Using AI:**

- Implement machine learning to classify normal vs. crash events using sensor data.

3. **GPS & GSM Alert System:**

- Send automatic SMS alerts with real-time location to emergency contacts.

4. **Cloud and Mobile App Integration:**

- Store ride and crash data in cloud and display it on a mobile dashboard.

5. **Computer Vision Module:**

- Use camera and OpenCV to verify helmet wearing and capture incidents.

6. **Data Analytics & Visualization:**

- Analyze ride patterns, accident frequency, and safety metrics using Power BI or Python.

7. **Scalability and Real-Time System:**

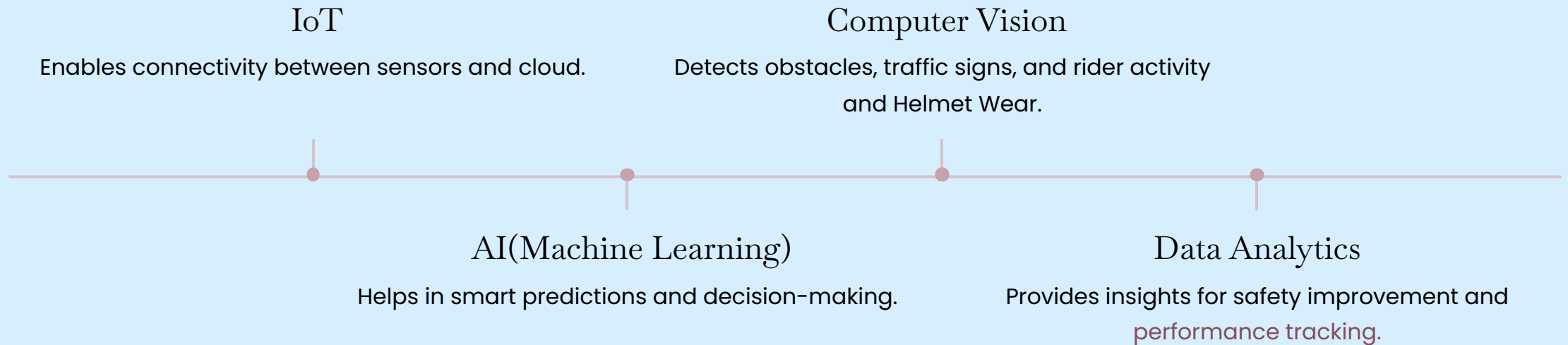
- Build a reliable, low-latency IoT system suitable for real-world use.

Methodology:

Our Smart Helmet integrates multiple advanced technologies to enhance rider safety and accident prevention.

It combines **IoT**, **Artificial Intelligence**, **Computer Vision**, and **Data Analytics** to collect, process, and analyze real-time data for intelligent decision-making.

Key Highlights:



Internet of Things (IoT):

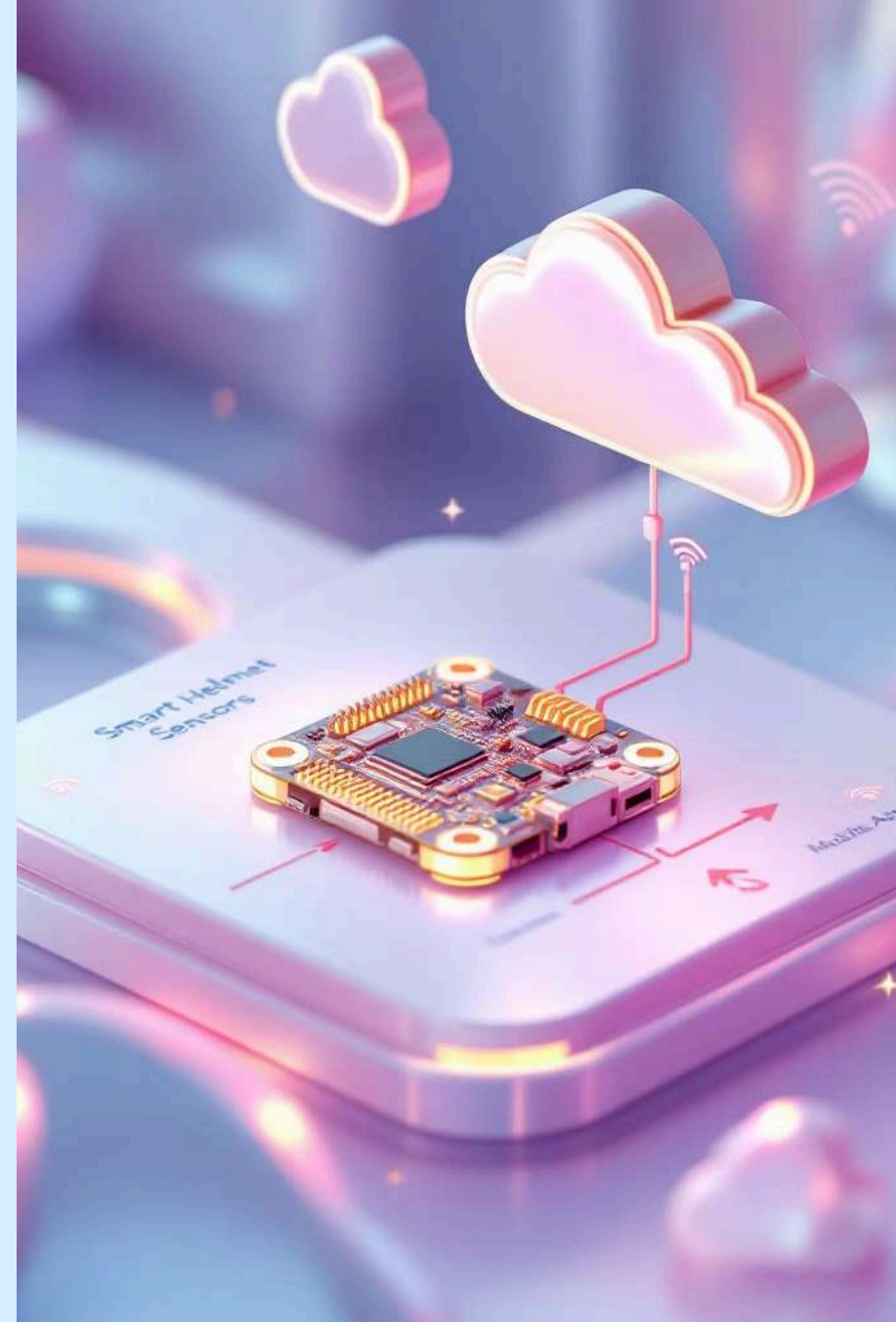
IoT (Internet of Things) connects the Smart Helmet's hardware components sensors, microcontroller, and cloud – to communicate and share real-time data. This connectivity allows continuous monitoring of the rider's environment and health, and immediate alerts in case of an accident.

Key Functions:

- **Sensor Integration:** Collects data from accelerometer, gyroscope, and GPS sensors.
- **Microcontroller Communication:** Processes sensor data and sends it to the cloud via GSM module.
- **Real-Time Alerts:** Transmits accident alerts or abnormal conditions to the rider's phone and emergency contacts.
- **Cloud Connectivity:** Stores helmet data for further analysis and visualization.

Example Hardware:

- **Sensors:** MPU6050 (Gyroscope + Accelerometer), GPS Module.
- **Microcontroller:** ESP32 or Arduino
- **Communication Protocols:** MQTT, HTTP, or Blynk IoT



Computer Vision in Smart Helmet:

Computer Vision (CV) enables the Smart Helmet to “see” and understand visual data using a camera and AI algorithms. It enhances safety by detecting whether the rider is wearing the helmet and monitoring surroundings.



Helmet Detection

Detects if the rider is wearing the helmet using image recognition (YOLO / OpenCV).



Drowsiness Detection

Analyzes face and eyes to detect fatigue.



Accident Capture

Records and sends short video clips before and after a crash.



Traffic Vision

Detects nearby vehicles or obstacles using the camera feed.



Libraries & Tools

- OpenCV – for image capture and processing
- TensorFlow / PyTorch – for training deep learning models (object & face detection)
- YOLO or Haar Cascades – for real-time object detection

AI for Smart Decision-Making and Accident Prediction:



Smart Decision-Making

Artificial Intelligence enables the Smart Helmet to **think, learn, and make intelligent decisions** based on data from sensors and cameras. By analyzing motion patterns, head movement, and surrounding visuals, AI can detect risky behaviors and predict possible accidents before they happen.



Accident Prediction

Machine learning models analyze speed, tilt angle, and vibration to detect sudden impact or collision.



Anomaly Detection

AI identifies abnormal motion or drowsy driving patterns.



Safety Recommendations

The system learns from historical data to improve rider safety over time.



Machine Learning Algorithms

Utilizes Decision Trees, SVM, or Neural Networks for pattern recognition.



Training Data

Sensor readings (accelerometer, gyroscope, GPS) and video input from camera are used for training.

Data Analytics for Insights and Safety Improvements:

Data Analytics plays a vital role in transforming the raw data collected by the Smart Helmet into meaningful information. It helps identify accident patterns, risky behaviors, and system performance to enhance safety features over time.

Key Functions:

- **Data Collection:** Real-time data from sensors (speed, tilt, impact, GPS) and camera feed are stored in the cloud.
- **Data Cleaning & Processing:** Removes noise and organizes data for analysis using Python libraries like pandas and numpy.
- **Trend Analysis:** Detects frequent accident zones, average speed, and helmet usage duration.
- **Predictive Analytics:** Forecasts potential risk areas or unsafe conditions based on historical data.
- **Dashboard Visualization:** Displays live stats and trends through graphs and dashboards (e.g., using Power BI or Matplotlib).

Tools & Libraries:

- **Python (pandas, numpy, matplotlib, seaborn)** – for analysis and visualization
- **Power BI** – for dashboards and reports
- **Cloud Database (Firebase / Thingspeak / AWS)** – for storing IoT data

How All Technologies Work Together in the Smart Helmet:

The Smart Helmet is a fully integrated system where **IoT, AI, Computer Vision, and Data Analytics** work together to ensure rider safety, data-driven insights, and intelligent decision-making. Each technology plays a unique role but operates in coordination to create a seamless ecosystem.



IoT Layer

Sensors collect real-time data (speed, tilt, GPS, heart rate) and send it to the microcontroller.



AI Layer

The AI model analyzes patterns to detect possible accidents or unsafe behavior.



Computer Vision Layer

The camera processes live video to detect obstacles, drowsiness, or road signs.



Data Analytics Layer

Collected data is sent to the cloud for analysis, visualization, and safety optimization.



User Interface

Alerts, notifications, and reports are sent to the rider's smartphone or dashboard.

Cloud Integration & Data Management:

The cloud platform acts as the **central hub** for storing, processing, and analyzing all data collected from the Smart Helmet. It connects the IoT device, mobile app, and data analytics layer to ensure *real-time synchronization* and *secure data flow*.

Key Functions:

- 1. **Data Storage:**
 - Saves sensor readings (speed, tilt, GPS, impact) in the cloud database.
 - Stores accident logs, user profiles, and alert history.
- 2. **Real-Time Sync:**
 - Updates data instantly between the helmet and mobile app using *Firebase Realtime Database* or *Firestore*.
- 3. **Crash Alert Management:**
 - Cloud triggers push notifications or sends alerts to contacts through integrated APIs.
- 4. **Analytics & Dashboard:**
 - Data from the cloud is analyzed using *Python / Power BI* for pattern detection and safety reports.
- 5. **Authentication & Security:**
 - Firebase Auth ensures that only authorized users can access helmet data.
 - SSL encryption secures all data transfers.

Technologies Used:

Purpose	Tool/Service
Cloud Database	Firebase Realtime DB / Firestore
Cloud Storage	Firebase Storage / AWS S3
Notifications	Firebase Cloud Messaging (FCM)
Hosting	Firebase Hosting / Node.js API
Analytics	Python, Power BI, Thingspeak

4. GPS Module (NEO-6M) – Pinpointing the Emergency:

In the critical moments following an accident, knowing the precise location of the rider can dramatically reduce response times and improve outcomes. The GPS module is essential for this life-saving function.



Satellite-Based Tracking

The NEO-6M GPS module connects with orbiting satellites to triangulate and determine its exact position on Earth, providing highly accurate latitude and longitude coordinates.



Integration with Emergency Alerts

The acquired GPS coordinates are then embedded directly into the emergency SMS message. This vital information allows rescuers to quickly navigate to the accident site, even in remote or unfamiliar areas.

The NEO-6M module's reliability and compact size make it an excellent choice for integration into the Smart Helmet, ensuring that help can find the rider swiftly when it matters most.



Real-Time Location Data

Upon accident detection by the accelerometer and gyroscope, the ESP32 microcontroller interfaces with the GPS module to obtain the current location in real-time, ensuring the most up-to-date position is transmitted.



Enhancing Response Efficiency

By eliminating the need for accident witnesses or victims to communicate their location, the GPS module significantly streamlines the emergency response process, potentially saving crucial minutes or even hours.

GSM Module (SIM800L/SIM900) – Automated Emergency Communication:

The GSM module is the communication backbone of our Smart Helmet's emergency response system, enabling it to reach out for help autonomously when a serious accident is detected.



Cellular Connectivity

Operating like a miniature mobile phone, the GSM module utilizes a standard SIM card to connect to cellular networks, allowing it to send SMS messages and potentially make calls.

Automated SMS Dispatch

Upon confirmation of an accident by the ESP32 microcontroller, the GSM module is instructed to send pre-configured emergency SMS messages to multiple stored contact numbers.

Location-Enabled Alerts

Crucially, the GPS coordinates obtained from the NEO-6M module are dynamically embedded into the outgoing SMS. This provides recipients with immediate and actionable location information.

Reliable Emergency Contact

This system ensures that designated contacts—family, friends, or emergency services—are instantly notified, even if the rider is incapacitated and unable to call for help themselves.

The selection of modules like SIM800L or SIM900 is based on their proven reliability, widespread network compatibility in regions like Pakistan, and cost-effectiveness, making emergency communication accessible and dependable.

Mobile Application for Smart Helmet:

Helmet-App Connectivity

- Helmet uses **ESP32** with built-in **Bluetooth & GSM/Wi-Fi** to connect with the app.

Accident data (location, impact force) is automatically shared with the mobile app.

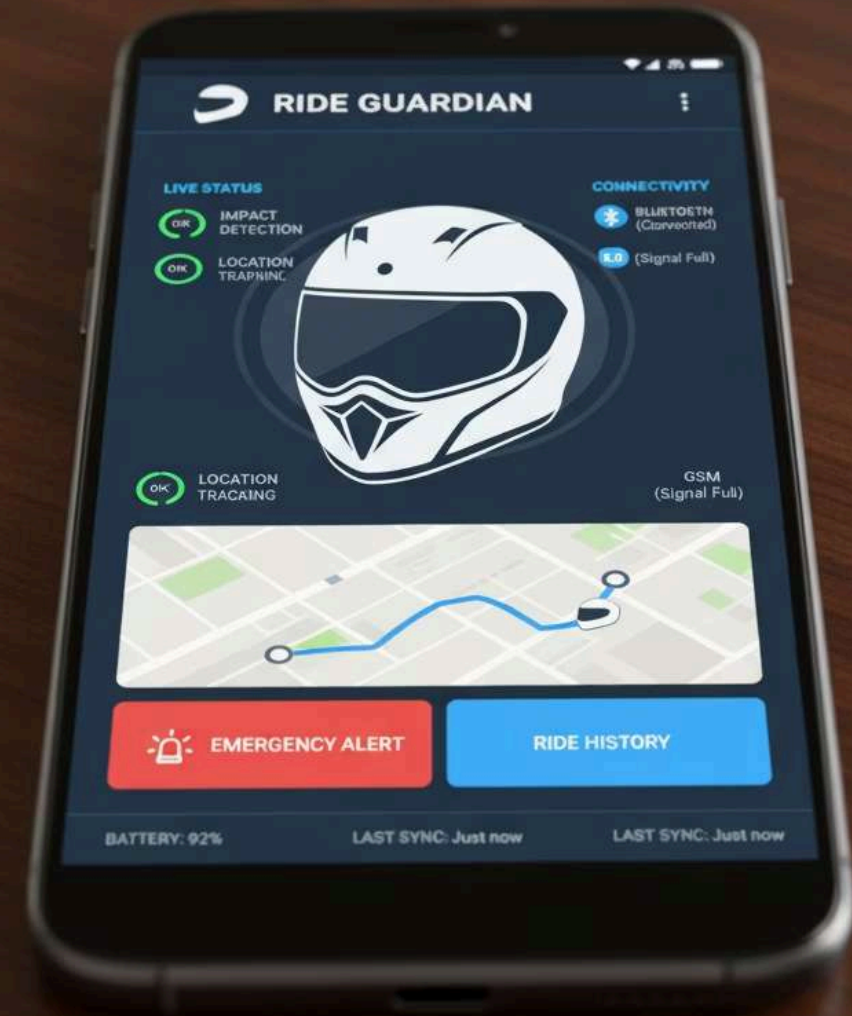
App Features

- **Helmet & Bike Status** → Connected/Not Connected.
- **Live Location Tracking** → GPS with Google Maps integration.
- **Accident Alerts** → Emergency SMS + App Notification.
- **Ride History** → Distance, speed, past alerts.

Settings → Add/edit emergency contacts, sensitivity control.

UI Design Highlights

- Clean, simple, and emergency-focused.
- Black + Yellow theme (safety color scheme).
- **Large buttons, clear alerts, rider-friendly layout.**



How the Smart Helmet System Works (Step-by-Step Process)



Helmet Detection

- When the rider wears the helmet, it activates.
- The system verifies the presence of the helmet before allowing bike ignition.



Ignition Control

- Arduino checks helmet status; if helmet is not detected, relay module blocks ignition.
- Once verified, the rider can start the bike safely.



Sensor Monitoring

- During the ride, sensors (MPU6050, GPS, etc.) continuously monitor acceleration, tilt, and position.
- Data is sent to the microcontroller in real time.



Accident Detection

- Sudden impact or abnormal acceleration triggers the AI/ML model to classify it as a potential accident.



Alert & GPS Tracking

- Upon crash detection, GPS module collects coordinates.
- GSM module sends SMS with location to emergency contacts.



Cloud Update & Analytics

- All sensor and event data are uploaded to Firebase Cloud.
- Data is analyzed later using Python and Power BI to find patterns and improve safety insights.



Mobile App Notification

- The connected mobile app displays real-time data, crash alerts, and map view of current location.

"Our Smart Helmet works as a complete safety ecosystem. It ensures helmet use, detects crashes through AI, sends alerts via GSM and GPS, and stores all data in the cloud for analysis and reporting."

Conclusion And Future Enhancements:

Conclusion:



Integrated Safety System

The **Smart Helmet System** successfully integrates **IoT, Artificial Intelligence, Computer Vision, and Data Analytics** to improve rider safety and accident response.



Real-time Monitoring & Alerts

It ensures that the rider **wears the helmet**, monitors **real-time motion and crash data**, and sends **immediate alerts** with location through GSM/GPS.



AI-Powered Detection

Using **AI models**, it detects accidents accurately, while **computer vision** verifies helmet wearing and road safety conditions.



Valuable Insights

Data analytics provides valuable insights such as rider behavior, accident hotspots, and performance trends.



Smart Infrastructure Contribution

This system not only enforces safety but also contributes to building **smart transportation infrastructure**.

Future Enhancements:



Edge AI Processing

Real-time crash prediction directly on the device without internet.



Solar-Powered Helmet

For extended battery life and energy efficiency.



Voice Assistant Integration

For hands-free control (Google Assistant / Alexa).



Advanced Cloud Dashboard

AI-based insights, accident heatmaps, and predictive analytics.



Emergency Services API Integration

Automatic dispatch to nearest hospitals.



Fleet Management System

For delivery riders and companies.

References:

1. Espressif Systems. *ESP32 Technical Reference Manual*. Available at: <https://www.espressif.com/en/products/socs/esp32>
2. Texas Instruments. *Accelerometer & Gyroscope Sensors – Application Notes*. Available at: <https://www.ti.com/sensors/motion/overview.html>
3. IEEE Xplore. *Smart Helmet Systems for Accident Detection and Prevention*. Available at: <https://ieeexplore.ieee.org>
4. Arduino Official Docs. *Using RFID for Authentication Projects*. Available at: <https://docs.arduino.cc>
5. World Health Organization (WHO). *Global Status Report on Road Safety 2023*. Available at: <https://www.who.int/publications/i/item/9789240061537>
6. ResearchGate. *Accident Detection and Alerting System Using IoT and Sensors*. Available at: <https://www.researchgate.net>

THANK YOU!